Lakshana Tekula

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EDUCATION

The University of Memphis **Jan 2023 – Dec 2024**
Master of Science in Data Science *CGPA : 3.9*

EXPERIENCE

First Horizon **Sep 2024 – Dec 2024**
Data Architect Intern *Memphis, TN*

- Assisted with data collection, preprocessing, and migration to Azure Cloud, reducing infrastructure costs by 12%.
- Spearheaded integration of GenAI technologies into data architecture, enhancing predictive analytics and automation, which improved decision-making processes
- Participated in Architecture Review Board (ARB) meetings to align data strategies with organizational goals and ensure scalability and robustness in Machine Learning systems.

Infosys Limited **Jan 2022 – Dec 2022**
Machine Learning Engineer *Hyderabad, India*

- Developed and deployed Python based training pipelines using Azure cognitive services for sentiment analysis, text classification and entity recognition, improving customer satisfaction by 22%.
- Containerized NLP models using Azure Kubernetes Service for scalable and seamless integration into client systems.
- Leveraged clustering algorithms to segment customers based on behavior, informing the development of recommendation systems and predictive models that increased marketing campaign conversion rates by 20%.
- Initiated a comprehensive approach to model explainability, with a focus on data insights and a calibrated combined metric, resulting in improved model efficency and model understanding for end-users.
- Collaborated with cross-functional teams on client projects to integrate Named Entity Recognition (NER) models with custom dictionaries, enhancing entity extraction by 15% for industry-specific terminology.

MTAR Technologies Limited **July 2020 – Jan 2022**
Jr. Machine Learning Engineer *Hyderabad, India*

- Conceptualized Business model by Data Collection, Data Modeling, Data Exploration and developing Dashboard to see current trends and give Logical Recommendation using R, SQL and PowerBI which improved the success rate by 25%.
- Reduced cycle times by 15% through in-depth analysis of production data, leveraging statistical techniques such as hypothesis testing to identify key correlations and patterns.
- Forecasted demand with 95% accuracy using models like XGBoost, reducing inventory holding costs by 12% and optimizing supply chain operations to align with production schedules.
- Engineered and fine-tuned CNN architectures (ResNet, MobileNet) for defect detection and quality control, achieving a 10% improvement in defect identification accuracy and minimizing false positives on the production line.
- Utilized U-Net and DeepLab models for high-precision segmentation, supporting defect localization and dimensional measurement tasks, leading to enhanced product quality and adherence to specifications.

JNTUHCEH **Jan 2020 – Jul 2020**
Research Assistant *Hyderabad, India*

- Optimized SQL databases, ensuring efficient organization of research data while implementing indexing strategies and schema enhancements to reduce query response time by 18% and improve data retrieval speed for analytics.
- Implemented unsupervised machine learning techniques, such as DBSCAN and Isolation Forest, for time-series anomaly detection on 1TB of unstructured data, reducing processing time by 20%.

PROJECTS

Advanced Question Answering System | *NLP, LLMs, Transformers* **Feb 2024**

- Integrated state-of-the-art NLP models like BERT and RoBERTa for text preprocessing, named entity recognition, and reading comprehension, supplemented by rule-based systems for answer generation. Incorporated a feedback loop to continuously refine the models based on user interactions.

Credit Card Fraud Detection System | *Scikit-learn, Deep Learning, TensorFlow* **Nov 2023**

- Designed a comprehensive solution integrating Auto-Encoder (AE) modeling and t-SNE for visualizing transaction patterns, enabling better detection and interpretation of anomalies in financial datasets.

TECHNICAL SKILLS

Programming Languages: Python, Java, R, SAS, SQL
Visualization Tools: PowerBI, Tableau, Excel, Matplotlib, Seaborn, GGplot2, Plotly, Atlair
Big Data & Cloud: GitHub, Hadoop, Spark, Hive, AWS Bash, AWS SageMaker, Azure AI Services, Azure ML Studio
Statistics:: A/B testing, Statistical Inference, Causal Impact Analysis
AI/ML Systems: TensorFlow, PyTorch, Keras, Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Decision Trees, Reinforcement Learning, XGBoost, Sci-kit Learn, CNNs, GANs